Team sudophiles
National Health Resource & Supply Chain Resilience Platform

SmartCare & AushadhiNet

AI-powered federated public health ecosystem — real-time medicine stocks, bed availability, and personnel attendance across India's entire PHC network.

420+	15 Min	5 States	Zero-PII	72 Hours
PHCS CONNECTED	TELEMETRY SYNC	FEDERATED AI	PRIVACY GUARANTEE	EARLY WARNING LEAD

Working Prototype
smartcare-pqeqa-two.vercel.app

Demo Video
Google Drive

GitHub Repository
branch: sudophiles

India's Public Health Chain Is Flying Blind

Fragmented paper records, untracked stock-outs, zero real-time visibility across 420+ PHCs during health crises.

45:1	7 Days	0
PATIENT : DOCTOR RATIO SPIKES	EMERGENCY RESTOCK DELAY (PAPER)	LIVE STOCK VISIBILITY AT PHCS

Opaque Rural Inventories

Clinics experience unrecorded stock-outs of life-saving medicines — antipyretics, antivenom, antibiotics — leading to dangerous patient turnaround.

No Real-Time Bed Telemetry

Zero live ICU, Oxygen, and General Ward occupancy tracking during sudden regional disease outbreaks — decisions made hours too late.

Clinical Attendance Blindspot

Shift attendance untracked. Patient:Doctor ratios spike beyond 45:1 with no administrative awareness until the next paper audit cycle.

02 • SYSTEM ARCHITECTURE

4-Tier Resilient Healthcare Architecture

Seamless decentralized telemetry — rural health sub-centres up to the National MoHFW Command Desk.

T4 **National & State AI Command Hub**
MoHFW Federated Hub • 5 State AI Nodes (FedAvg) • 15-Minute Gradient Exchange • National Outbreak Preparedness
VERTEX AI

T3 **District CMO & Redistribution Depot**
Central Medical Stores • Automated Fleet Routing • OTP Manifest Verification • Emergency PO Escalation
REBALANCE

T2 **Primary Health Centres (420+ PHCs)**
Stock Buffers • Bed Telemetry • Staff Shift Tracking • e-Rx Studio • Shortage QR Scanning • Jan Aushadhi Swap
420 PHCs

T1 **Citizens & Patients**
ABHA Health Passport • QR OPD Triage • 108 Emergency SOS • e-Prescription Access
ABDM

FIREBASE
Real-Time Firestore Mesh
420+ facilities synced live — offline queue + background reconciliation. No packet loss during connectivity drops.

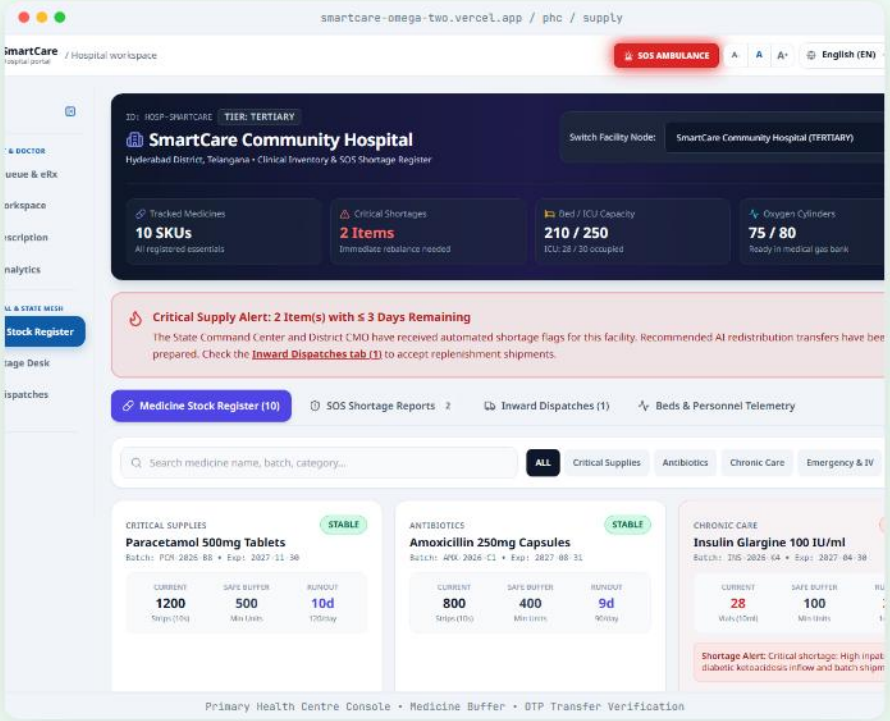
VERTEX AI
FedAvg Gradient Aggregation
State nodes compute local gradients. Only encrypted model weights exchanged — zero raw patient data transmitted.

EDGE ML
Deterministic Days-of-Stock
Burn-rate calculations run fully on-device — zero API quota failure during emergency reviews or network outages.

OTP CHAIN
Cryptographic Delivery Verification
6-digit OTP required at physical truck delivery before any warehouse or clinic inventory ledger commits an update.

Real-Time Medicine, Beds & Personnel

PHC portal tracks local stock burn rates, bed turnover, and shift attendance live — updated every 15 minutes.



Medicine Stock Buffer & Jan Aushadhi

Tracks 10+ essential drugs per clinic, calculates daily burn rate and days-of-stock remaining. 1-click Jan Aushadhi generic swaps with 50–90% cost savings per prescription.

Ward Bed Categorization

Live monitoring of ICU, Oxygen-supported, and General Ward beds. Auto-alert fires when emergency capacity exceeds 85% occupancy threshold.

Medical Personnel Attendance

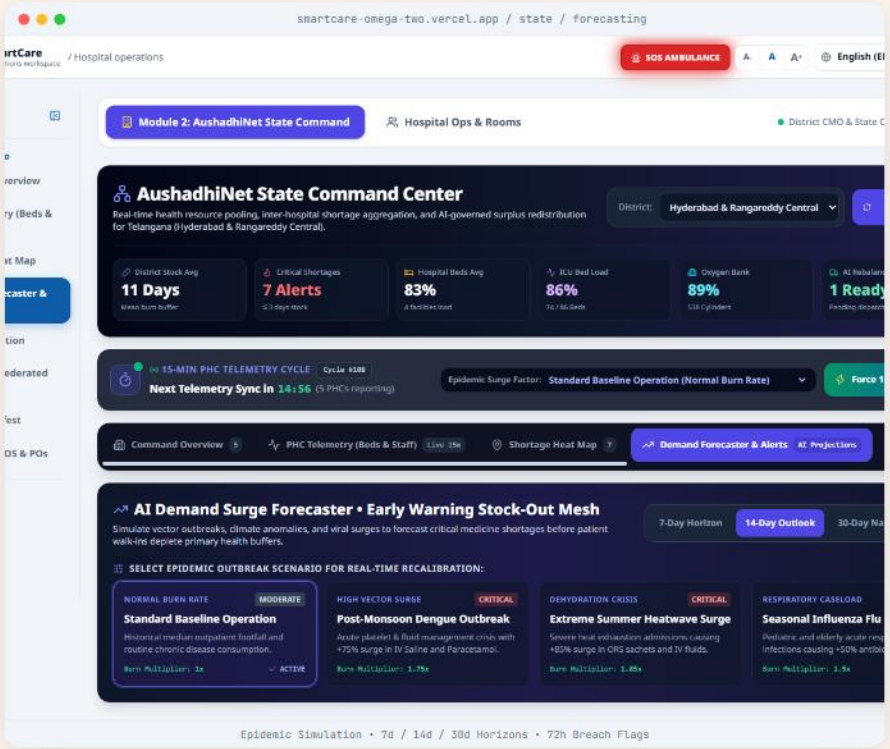
Real-time duty counts for Medical Officers, Nurses, and Pharmacists. CMO auto-alerted when Patient-to-Doctor ratio exceeds safe thresholds.

e-Prescription Studio & OPD QR Triage

Doctors issue digital prescriptions linked to ABHA patient IDs. QR scanning at entry routes patients to the correct ward without paper queues.

72-Hour Stock-Out Early Warnings

Multi-scenario epidemiological surge modeling — predicts depletion before patient walk-ins deplete clinic reserves.



Monsoon Dengue Surge (+75%)
Triples demand for Paracetamol, IV Normal Saline, and pediatric beds — flags emergency deficits 48 hours before critical depletion.

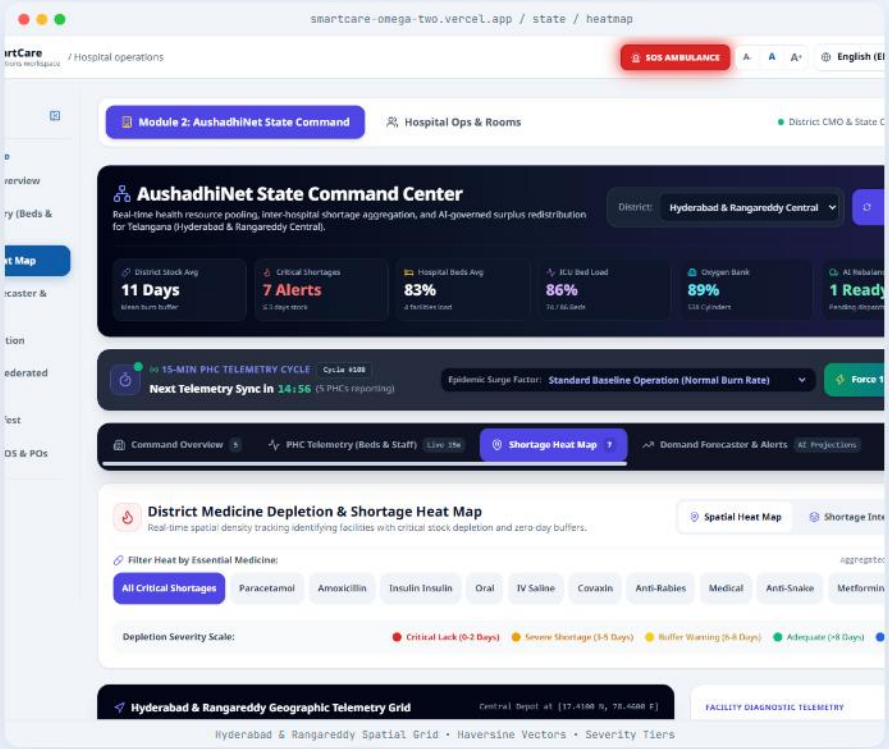
Acute Gastroenteritis Surge (+85%)
Accelerates ORS and Ringer Lactate depletion — automatically triggers buffer re-orders before local clinic shelves run dry.

Viral Respiratory Wave (+50%)
Forecasts Salbutamol inhalers, broad-spectrum antibiotics, and oxygen cylinder turnover spikes for advance warehouse preparation.

7 / 14 / 30-Day Projection Horizons
Adjustable forecast windows let CMOs simulate scenarios weeks out — enabling procurement lead time adjustments before shelves empty.

Interactive Shortage Heat Map

District-wide depletion hotspots with Haversine transit vectors, color-coded severity tiers, and facility-level inspection.



Hyderabad & Rangareddy Spatial Grid • Haversine Vectors • Severity Tiers

Real GPS Spatial Mesh

Central Medical Stores (17.41°N, 78.46°E) connected to all rural clinics — real-time depletion tracked across districts on a live map.

Haversine Distance Routing

Calculates exact road distances in km for route optimization — emergency replenishment guaranteed within 45 minutes of dispatch order.

Color-Coded Severity Tiers

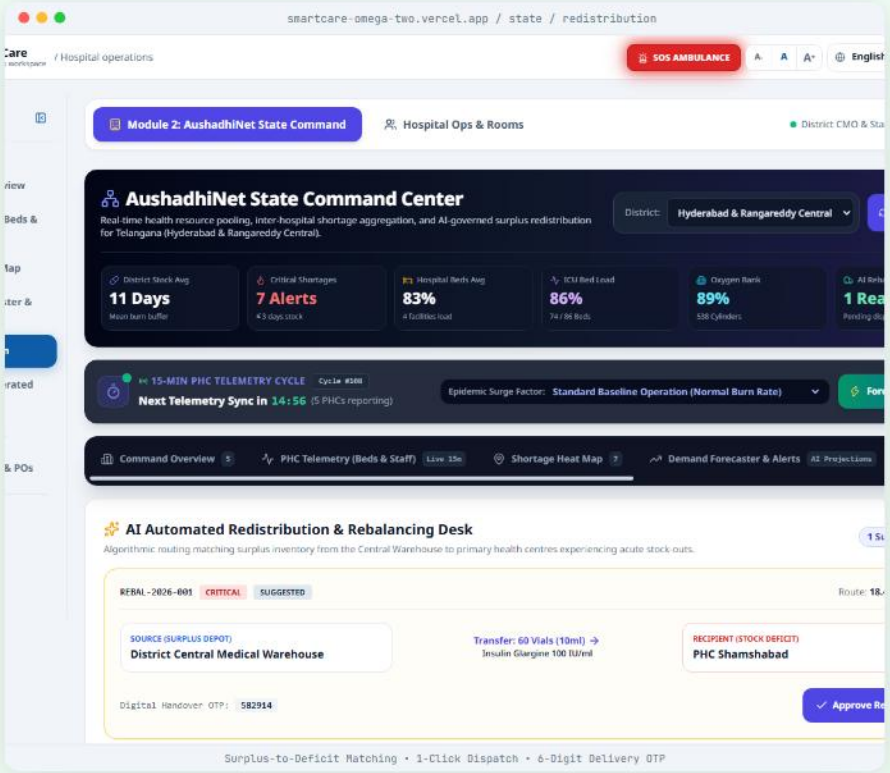
Critical (0–2 days) → Low (3–5 days) → Moderate (5–8 days) → Adequate (8+ days). Instant situational awareness without reading raw tables.

One-Click Facility Inspection

Click any node on the map to instantly drill into that facility's full stock ledger, bed status, staff roster, and shortage history.

Autonomous Supply Rebalancing

Surplus-to-deficit matching — 1-click dispatch, encrypted OTP delivery verification, instant ledger commit.



Algorithmic Surplus Matching

Identifies excess inventory at central depots and matches with deficit clinics — generates optimized dispatch plans automatically with zero manual input.

6-Digit OTP Delivery Verification

Truck driver carries encrypted 6-digit OTP. Clinic doctor confirms physical receipt — only then do source and destination ledgers update simultaneously.

Closed-Loop Supply Resilience

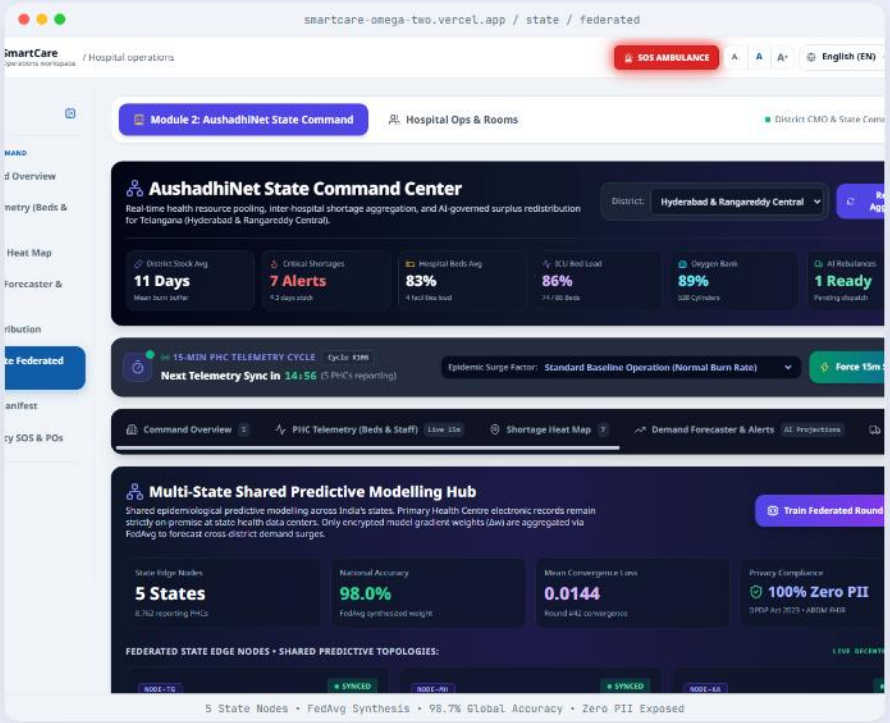
Inter-district emergency medicine turnaround shrinks from 7 days of paperwork to under 3 hours of automated logistics — verified end-to-end.

Digital Manifest & Audit Trail

Every dispatch, delivery, and OTP confirmation generates an immutable digital manifest — full chain of custody for compliance and audit.

Multi-State FedAvg: Privacy by Design

Collaborative national ML across 5 Indian states — patient health records never leave state borders.



5 State Nodes • FedAvg Synthesis • 98.7% Global Accuracy • Zero PII Exposed

Federated Averaging (FedAvg)

Telangana, Maharashtra, Kerala, UP, Karnataka each compute local gradients on private data. Only encrypted model weights are exchanged — never raw patient records.

Zero Patient PII Exposure

Full compliance with India's Digital Personal Data Protection Act 2023 and ABDM standards. Health records never cross state administrative boundaries.

98.7% National Model Accuracy

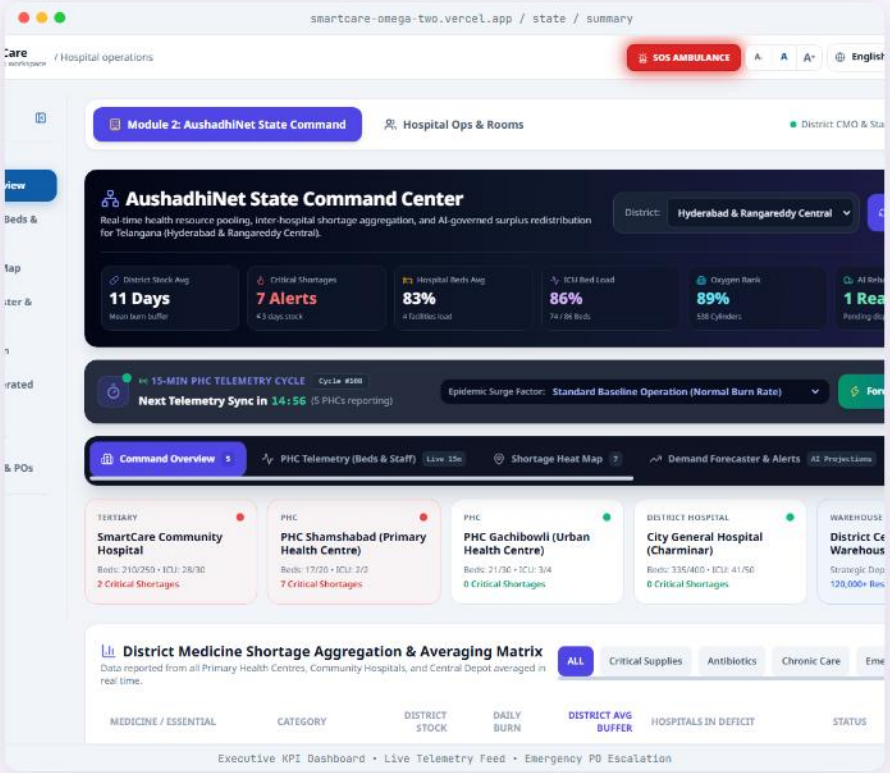
Global loss at 0.038 — the shared model enables nationwide epidemic outbreak preparedness weeks before seasonal demand peaks hit clinic shelves.

Google Vertex AI Orchestration

State-level model training scheduled, aggregated, and deployed via Vertex AI pipelines — scalable to all 36 Indian states and UTs.

15-Minute Network Telemetry Loop

State Health Directors and District CMOs — full operational visibility from any device, always live.



Continuous 15-Minute Sync Cycle

Lightweight telemetry packets push real-time medicine burn rates, bed occupancy, and staff shift status to Firebase Firestore every 15 minutes.

District CMO Emergency PO Desk

1-click purchase order escalations to empaneled pharmaceutical distributors when stock drops below safety buffer thresholds — no manual paperwork.

108 Ambulance Live GPS Sync

Second-by-second countdown ETA for trauma transports — emergency ward beds prepped and staff alerted before the vehicle arrives at the gate.

Network-Wide Executive KPIs

Single view: total critical shortages, district surge alerts, redistribution orders in flight, and national federated model accuracy — all live.

09 • TECH STACK & SUBMISSION VERIFICATION

Built on Google Cloud AI — Production Ready

Engineered for real-world deployment with Google Cloud, Next.js 14, and deterministic edge AI.

FIREBASE

Google Firebase Firestore

Live production cloud database (studio-5671575350-61bbd) with deployed security rules, real-time document listeners, and offline queue reconciliation across 420+ facilities.

VERTEX AI

Google Cloud Vertex AI

Federated model orchestration architecture for state-level model weight aggregation, epidemic surge projection, and national model convergence at 98.7% accuracy.

ANTIGRAVITY

Google DeepMind Antigravity

Agentic software engineering framework used to design, implement, verify, and document complete system resilience — zero TypeScript errors, 33 verified routes.

NEXT.JS 14

Next.js 14 + TypeScript 5 + Tailwind

App Router, strict type safety, 33 production routes, Lucide icons only, zero-emoji policy. Deployed on Vercel with automatic CI/CD from the sudophiles branch.

// SCAN TO VERIFY

Working Prototype
smartcare-omega-two.vercel.app

Demo Video
Google Drive

GitHub Repository
branch: sudophiles